DAY 07 OF 10

Modules, Files & Exceptions

Reuse libraries, read and write files and handle errors safely

Today's agenda

1. import and the standard library
2. pip and third-party packages
3. Reading and writing files
4. JSON and CSV
5. try / except / finally

Python in 10 Days
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Modules and import

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- A module is a .py file with reusable code
- import module or from module import name
- The standard library ships with Python: math, random, datetime, os, json, csv
- Your own file can be imported too (same folder)
- dir(module) lists what is inside; help(module) explains it

Python

```python
import math, random
from datetime import date

print(math.sqrt(49))
print(math.pi)
print(random.randint(1, 6))
print(date.today())
```

OUTPUT

```
7.0
3.141592653589793
4   (random)
2026-10-11   (today)
```

Installing packages with pip

- pip downloads third-party libraries from PyPI
- Install in a terminal, then import in your code
- Use a virtual environment so projects stay separate
- requirements.txt lists the packages a project needs

Python

```
pip install requests pandas

# virtual environment
python -m venv venv
venv\Scripts\activate     # Windows
source venv/bin/activate  # Mac/Linux
pip freeze > requirements.txt
```

On Colab, put an exclamation mark first: !pip install package_name

Reading and writing text files

- open(path, mode): "r" read, "w" write (overwrites), "a" append
- Always use with: it closes the file for you
- Loop over the file object to read line by line
- Use encoding="utf-8" for non-English text

Python

```python
with open("notes.txt", "w") as f:
    f.write("Line one\n")
    f.write("Line two\n")

with open("notes.txt") as f:
    for line in f:
        print(line.strip())
```

OUTPUT

```
Line one
Line two
```

Working with JSON

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- JSON is the standard format for web data
- json.dump writes a Python dict/list to a file; json.load reads it back
- json.dumps / json.loads work with strings
- Maps cleanly to dicts, lists, str, numbers, True/False, None

Python

```python
import json

data = {"name": "Asha", "skills": ["SQL", "Python"]}
with open("data.json", "w") as f:
    json.dump(data, f, indent=2)

with open("data.json") as f:
    loaded = json.load(f)
print(loaded["skills"])
```

OUTPUT

```
['SQL', 'Python']
```

Working with CSV

- CSV = comma separated values, the spreadsheet export format
- csv.DictReader gives each row as a dictionary
- csv.writer writes rows
- For big analysis, pandas (Day 9) is easier

Python

```python
import csv

with open("students.csv", newline="") as f:
    for row in csv.DictReader(f):
        print(row["name"], row["marks"])

with open("out.csv", "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(["name", "marks"])
    w.writerow(["Asha", 82])
```

Exceptions: handling errors

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- Errors at run time raise exceptions and stop the program
- try / except catches them so the program continues
- else runs when no error happened; finally always runs
- Catch specific exceptions, not everything

Python

```python
try:
    n = int(input("Enter a number: "))
    print(100 / n)
except ValueError:
    print("Please enter digits only")
except ZeroDivisionError:
    print("Cannot divide by zero")
else:
    print("Success")
finally:
    print("Done")
```

OUTPUT

```
Enter a number: 0
Cannot divide by zero
Done
```

Common errors and raising your own

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Error	Why it happens	Example
SyntaxError	Invalid code structure	print("hi"
NameError	Variable not defined	print(x)
TypeError	Wrong type for operation	"a" + 1
ValueError	Right type, bad value	int("abc")
IndexError	List index out of range	[1, 2][5]
KeyError	Dict key missing	d["zzz"]
FileNotFoundError	File does not exist	open("no.txt")

> Raise your own: raise ValueError("Age cannot be negative")

Practice time

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1. Write a program that stores 5 names in a file and reads them back
2. Count the words in a text file and print the 5 most common
3. Ask for a number and keep asking until the user enters a valid one (use try/except)
4. Read a CSV of marks and print the class average
5. Save a dictionary to JSON, change it, and save again

Challenge: a simple expense tracker that appends entries to a CSV file

Day 7 recap

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- import brings in modules; pip installs packages
- with open(...) for safe file handling
- json and csv are the two data formats you will meet most
- try / except / else / finally keep programs from crashing

Homework: finish practice. Next: Object Oriented Programming